Airway Smooth Muscle-Specific Transcriptomic Signatures of Glucocorticoid Exposure

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ONLINE DATA SUPPLEMENT

Online Supplement

Detailed Methods

ASM Cell Culture and Treatment

Primary ASM cells were isolated from white non-smoking donors, 9 who died of fatal asthma and 8 with no chronic illness or medication use. ASM cell cultivation was as described previously (1, 2). Briefly, passages 3 to 4 ASM cells maintained in Ham's F12 medium supplemented with 10% FBS, $CaCl_2$, buffered with HEPES, penicillin/streptomycin, primoch, and additional L-glutamine were used in all experiments. The F12 medium was used for culture because it provides Ca^{2+} levels that are consistent with seeing contractility of muscles in that media. Cells from each donor were treated with 100 nM budesonide or control vehicle (DMSO) for 24 hours.

RNA-Seq Library Construction and Sequencing

Total RNA was extracted from cells using the miRNAeasy mini kit (Qiagen Sciences, Inc., Germantown, MD). Using approximately 1 μg of RNA from each sample, the Illumina TruSeq Stranded mRNA LT Sample Prep Kit (Illumina, Inc., San Diego, CA) was used to prepare poly(A)-selected stranded RNA-Seq libraries, where second-strand cDNA synthesized with dUTP was digested, and only first-strand cDNA existed in the library. Ambion External RNA Controls Consortium (ERCC) RNA Spike-In Control Mix 1 (Life Technologies Corporation, Carlsbad, CA) was added to the samples. Amplified cDNA fragments were analyzed using the 2100 Bioanalyzer (Agilent Technologies, Inc., Santa Clara, CA) to determine fragment quality and size. Library concentrations were determined by Qubit Fluorometric Quantitation (Life Technologies Corporation, Carlsbad, CA). An Illumina HiSeq 2500 instrument was used to

generate 125 bp paired-end reads at the University of Pennsylvania Next-Generation Sequencing Core (Philadelphia, PA).

RNA-Seq Data Analysis

Taffeta scripts were used to analyze RNA-Seq data and generate HTML summary reports (https://github.com/blancahimes/taffeta) (3, 4). Adapter sequences in raw reads were trimmed out using Trimmomatic (v.0.32) (5), and overall QC metrics were obtained using FastQC (6) (v.0.11.7). Trimmed reads for each sample were aligned to the reference hg38 genome and known ERCC transcripts using STAR (7) (v.2.5.2b), while constraining mapped reads to be within reference hg38 or ERCC transcripts. Bamtools (v.2.3.0) (8) was used to summarize the number of mapped reads, including junction spanning reads. The Picard Tools (v.1.96; http://picard.sourceforge.net) RnaSeqMetrics function was used to compute the number of bases assigned to various classes of RNA, according to an hg38 refFlat file obtained via a UCSC Genome Table. HTSeq (v.0.6.1, with the strand setting '-s reverse') was used to quantify both ERCC Spike-In and gene-level read counts, using provided hg38 human transcriptome and ERCC reference files (9). ERCC Spike-in dose-response curves (i.e., plots of ERCC transcript read counts vs. known quantity of ERCC transcript molecules added to samples) were created following the manufacturer's protocol (10). Gene-level differential expression analysis that compared (1) budesonide vs. control in non-asthma donor-derived ASM, (2) budesonide vs. control in fatal asthma donor-derived ASM, (3) fatal asthma vs. non-asthma donor-derived ASM exposed to control, and (4) fatal asthma vs. non-asthma donor-derived ASM exposed to budesonide, was performed under a negative binomial distribution model with DESeq2 (v.1.18.1) (11) while filtering out genes with a total read count <10 before differential analysis

and disabling the default option for independent filtering of differential expression results. Cell donor was used as a covariate in the differential expression analysis of glucocorticoid vs. control comparison; sex and age were used as covariates in fatal asthma vs. non-asthma donor-derived ASM comparisons. Ensembl gene IDs were annotated to official gene symbols using BioMart (2016 archive) (12). Genomic coordinates were retrieved from Ensembl Release 84 (March 2016). RNA-Seq data is available in GEO (https://www.ncbi.nlm.nih.gov/geo/) under accession GSE94335.

Gene Set Enrichment Analysis

Gene set enrichment analysis was performed with the fast gene set enrichment analysis algorithm implemented in the *fgsea* R package (v.1.4.1) (13). Gene sets of KEGG and Reactome pathway annotations were downloaded from MsigDB collections (v.6.2) (http://software.broadinstitute.org/gsea/msigdb/collections.jsp) and those with gene number >15 and <500 were included for analysis. Genes were sorted according to decreasing z-score from DESeq2 results. If multiple Ensembl gene IDs were annotated to the same gene symbol, that with the largest z-score was retained. Gene-level score for a gene in rank k was calculated by adding a positive or negative value, based on the presence or absence of the gene in the gene set, to the score of the gene in rank k-1, and positive values were weighted by the gene k's z-score. Enrichment score for a particular gene set was designated as the largest absolute value from all gene scores. P-values were assessed based on the distribution of enrichment scores following 10,000 permutations, and subsequently, q-values were obtained using the Benjamini-Hochberg approach. The leading-edge subsets, defined as genes in the ranked list that appear before and at the position of a positive gene-set enrichment score or that appear at and after the position of a

negative gene set enrichment score, correspond to genes that drove the enrichment signal. Dependent pathways with q-value <0.05 were collapsed using the *collapsedPathways* function in *fgsea*.

Analysis of Publicly Available Glucocorticoid-Response Transcriptomic Datasets

We sought publicly available transcriptomic datasets that measured the effect of glucocorticoid exposure on various human cell types to determine ASM-specific transcriptomic changes induced by glucocorticoids. GEO searches of the terms "*glucocorticoids*", "*budesonide*", "*dexamethasone*", and "*fluticasone*" in *Homo sapiens*, yielded 13 glucocorticoid response datasets (12 microarray and one RNA-Seq). Differential expression results of glucocorticoid vs. control comparison for individual studies were obtained using RAVED (https://github.com/HimesGroup/raved). For the RNA-Seq study, raw sequence data in *fastq* format was downloaded from SRA and taffeta scripts were used as described for the primary ASM studies above. For microarray studies, GSEMatrix and raw data files were downloaded from GEO. A quality control step obtained quality metrics of raw intensity data using outlier scoring methods from the *arrayQualityMetrics* R package (14). Scan date was considered as a batch variable via principal component analysis. Differential expression analysis was performed with the *limma* R package (15) using robust multi-array average (RMA)-processed intensity data (16). For studies with noticeable batch effects, the *sva* R package (17) was used to obtain adjusted statistics. The Benjamini-Hochberg approach was used to correct for multiple comparisons made. Microarray probes were annotated to official gene symbols using R array annotation packages corresponding to the appropriate platform. For cases in which multiple probe or Ensembl gene IDs annotated to the same official gene symbol, the smallest p-value was

retained for that gene symbol. Full differential expression results for each of these individual studies is available in our previously developed app REALGAR (http://realgar.org/) (18).

Tissue-Specific Integration of Glucocorticoid-Response Transcriptomic Data

For cell types that had more than one transcriptomic dataset, we obtained integrated results using two summary statistics-based approaches: (1) an effect size-based method that applied a random-effects model with heterogeneity of selected datasets estimated by DerSimonian and Laird's method (19), and (2) a rank-based method that adopted the Rank Product method (20). For (1), effect sizes measured as $\log_2$ fold change and standard errors from individual studies were passed to the *meta.summaries* function from the *rmeta* R package (21) to compute combined effect sizes and standard errors. Corresponding p-values were derived from resultant z-score distributions and converted to q-values using the Benjamini-Hochberg approach. For (2), genes present in each individual dataset were ranked according to their unadjusted p-values. Rank product statistics were computed using ranks re-scaled by the total number of genes. The approximate rank product distribution was obtained using 10^6 permutations, and empirical p-values were defined based on the proportion of simulated rank products that were smaller than the observed rank product. Corresponding q-values were conservatively estimated using false discovery rates. Gene significance per cell type was determined based on having a q-value <0.05 in both effect-size- and rank-based integration procedures. Full R code used for integration analysis is available at https://github.com/HimesGroup/Asthma_Transcriptomic_Analysis.

Hierarchical Clustering

Hierarchical clustering was performed to compare cell-type specific gene expression patterns across 15 glucocorticoid vs. control comparisons available among the 14 transcriptomic studies considered (including the comparisons for fatal asthma and non-asthma donor-derived ASM described above). The genes selected to perform clustering were 56 significantly differentially expressed genes in response to budesonide exposure in non-asthma ASM that also had results available in all datasets considered. Study and gene expression similarity were determined based on the Euclidean distance between $\log_2$ fold changes in expression as implemented in the *heatmap.2* function in the *gplots* R package.

Detailed RNA-Seq Quality Control Results

Thirty-four RNA samples from 17 ASM donors, 9 with fatal asthma and 8 non-asthma controls, with and without budesonide exposure, were obtained (Table E1). For these 34 samples, we obtained an average of 111.3 million trimmed sequencing reads per sample (range 87.9-312.0 million reads per sample). An average of 91.8% of RNA-Seq reads aligned to hg38 genome reference files downloaded from Illumina's iGenomes project (range 84.2%-94.7%), and an average of 34.6% of the mapped reads spanned junctions (range 31.9%-36.9%). An average of 94.1% of bases in mapped reads corresponded to mRNA (range 86.5%-95.3%). Plots of normalized read coverage of transcripts vs. normalized position revealed that there was even coverage of transcripts (Fig. E1A). ERCC spike-in dose response plots, which ideally would have slope and R^2 equal to 1.0, had an average slope and R^2 of 0.95 (range 0.92-0.99) and 0.94 (range 0.90-0.95) respectively (Fig. E1B shows the plot for a representative sample). Two housekeeping genes (*GABARAP* and *RPL19*) were used as positive controls for sequencing, and as expected, they were highly expressed viz-a-viz their normalized read counts and levels did not

differ significantly by either glucocorticoid exposure or disease status (Fig. E1C). Following the above quality control processing, all samples were deemed of sufficiently high quality to include in differential expression analyses.

Online Supplementary Tables

Table E1. RNA-Seq Quality Control Metrics. The table contains the cell line information, asthma and treatment condition, the number of raw reads for the paired-end samples after adaptor trimming (number of reads in both R1 and R2), the percentage of mapped reads among the total number of raw reads, the percentage of junction spanning reads among the mapped reads, the percentage of mapped bases that mapped to mRNA, the mean insert size of mapped reads, and the slope and R2 values of ERCC spike-in dose response curves for each sequenced samples. ERCC, Ambion External RNA Controls Consortium.

Sample	Cell Line	Asthma Status	Treatment	Total Raw Reads (Million)	Mapped Reads (Percent)	Junction Spanning Reads (Percent)	mRNA Bases (Percent)	Mean Insert Size (bp)	ERCC Spike-In Dose Response Curve (Slope)	ERCC Spike-In Dose Response Curve (R^2)
28788	AS013014	asthma	control	92.3	92.2	35.9	94.1	363.4	0.92	0.94
28797	AS013014	asthma	budesonide	87.9	92.0	33.8	87.7	255.9	0.96	0.94
28763	AS020811	asthma	control	141.5	93.6	33.7	94.7	371.9	0.96	0.93
28768	AS020811	asthma	budesonide	124.7	84.2	32.1	94.7	262.8	0.94	0.94
28762	AS021111	asthma	budesonide	98.3	93.1	35.9	94.7	379.5	0.94	0.94
28765	AS021111	asthma	control	122.0	92.9	33.9	94.4	344.5	0.98	0.94
28776	AS042913	asthma	control	93.3	90.1	34.5	95.2	306.0	0.96	0.94
28785	AS042913	asthma	budesonide	105.4	94.7	36.1	95.0	512.8	0.97	0.94
28786	AS090513	asthma	budesonide	89.7	92.6	33.3	86.6	260.4	0.94	0.94
28789	AS090513	asthma	control	101.8	92.1	32.3	86.5	277.1	0.95	0.95
28775	AS100614	asthma	control	93.6	92.1	34.9	95.3	329.7	0.94	0.93
28780	AS100614	asthma	budesonide	106.9	93.8	34.3	94.7	456.5	0.94	0.93
28787	AS110612	asthma	control	105.9	90.4	35.8	95.2	319.1	0.94	0.93
28792	AS110612	asthma	budesonide	110.5	88.3	34.3	94.4	314.4	0.97	0.95
28774	AS112514	asthma	budesonide	103.1	94.1	35.9	95.0	472.0	0.96	0.95
28777	AS112514	asthma	control	98.6	91.8	33.2	95.1	336.6	0.92	0.93
28764	AS113011	asthma	control	108.6	93.9	32.9	94.3	377.2	0.95	0.90
28773	AS113011	asthma	budesonide	110.7	90.6	36.0	95.0	424.7	0.97	0.93
28781	N010912	non-asthma	budesonide	99.0	93.2	35.8	94.9	346.1	0.96	0.94

28784	N010912	non-asthma	control	95.0	92.5	34.5	94.3	397.2	0.96	0.93
28779	N012412	non-asthma	control	108.3	93.6	33.3	94.6	323.0	0.93	0.91
28782	N012412	non-asthma	budesonide	24.0	91.9	35.4	94.5	362.4	0.94	0.92
28791	N021712	non-asthma	control	108.5	92.7	35.7	94.9	379.3	0.97	0.94
28794	N021712	non-asthma	budesonide	95.7	90.7	36.1	95.3	300.9	0.96	0.94
28769	N051911	non-asthma	budesonide	113.5	91.9	34.8	94.5	424.1	0.96	0.95
28772	N051911	non-asthma	control	109.9	89.6	31.9	94.2	404.2	0.94	0.92
28766	N061311	non-asthma	budesonide	108.0	91.9	34.3	94.6	291.7	0.95	0.93
28771	N061311	non-asthma	control	121.9	93.0	34.1	94.6	392.9	0.96	0.95
28790	N090710	non-asthma	budesonide	101.5	90.7	36.2	94.8	322.8	0.96	0.95
28795	N090710	non-asthma	control	109.1	92.4	35.7	94.8	402.8	0.96	0.94
28778	N092611	non-asthma	budesonide	312.0	93.5	36.9	94.7	556.6	0.99	0.95
28783	N092611	non-asthma	control	94.5	92.0	34.3	94.8	343.2	0.95	0.93
28767	N100811	non-asthma	control	106.8	91.2	34.4	94.6	389.7	0.96	0.93
28770	N100811	non-asthma	budesonide	109.1	91.2	33.1	94.1	268.3	0.94	0.94

Table E2. Summary of Evidence for Glucocorticoid Responsiveness-Related Roles of Differentially Expressed Genes. Top-ranked genes that have published reported evidence for playing a role in glucocorticoid response were selected from Tables 2 and 3.

Gene	Glucocorticoid Responsiveness-Related Evidence
ANGPTL7	Encodes a family of secretory proteins structurally similar to angiopoietins mRNA expression is up-regulated by dexamethasone in primary human trabecular meshwork cells (22) *ANGPTL7* knockdown during dexamethasone exposure significantly affects the expression of steroid-regulated extracellular matrix proteins (23)
CRISPLD2	Encodes serum protein implicated in lung development and endotoxin regulation (24) In ASM, mRNA expression is upregulated by dexamethasone or IL1β, and *CRISPLD2* knockdown increased the IL1β responsiveness of IL6 and IL8 (3) In ASM, transcription is induced by dexamethasone via glucocorticoid receptor occupancy (25) SNPs within the gene are moderately associated with inhaled corticosteroid resistance and bronchodilator response among asthma patients (3, 26)
FAM107A	Encodes a family of stress responsive small proteins mRNA expression is induced by dexamethasone in mouse hippocampus (27) and cultured astrocytes (28) In mouse liver cells, transcription is strongly induced by prednisolone dependent on glucocorticoid receptor dimerization (29)
FKBP5	Encodes a co-chaperone of hsp90 regulating glucocorticoid receptor sensitivity (30) *In vivo*, mRNA expression is up-regulated by inhaled budesonide of airways derived from non-asthmatic (31) and asthmatic subjects (32) *In vitro*, mRNA expression is up-regulated by glucocorticoids in ASM (3, 33, 34) In ASM, transcription is induced by dexamethasone via glucocorticoid receptor occupancy (25) In lymphoblastoid cell lines, mRNA expression is increased in asthma vs. non-asthma subjects (35)
GLUL	Encodes glutamine synthetase that converts glutamate to glutamine mRNA expression is up-regulated by dexamethasone in both human (36) and rat myocytes (37), and human ASM (3, 33, 34)

GPX3	mRNA expression is regulated independently via epigenetic or glucocorticoid receptor-mediated mechanisms in lung cancer cells (38)
ITGA10	mRNA expression is up-regulated by glucocorticoids in human airways *in vivo* (31) or *in vitro* (34), while is down-regulated in zebrafish larvae by prednisolone (39)
KLF15	Encodes one family of Kruppel-like, zinc finger transcription factors In ASM, mRNA expression is up-regulated by dexamethasone. Overexpression of *KLF15* reduces proliferation of human ASM cells, whereas apoptosis is increased in *KLF15* knockout murine ASM cells (33) In ASM, transcription is induced by dexamethasone via glucocorticoid receptor elements, and represses ASM hypertrophy (25)
PDK4	mRNA expression is up-regulated by dexamethasone in hepatoma cells (40) In rat primary hepatocytes, transcription is induced by dexamethasone via glucocorticoid receptor occupancy (41)
PER1	Encodes a transcription factor that regulates circadian rhythms mRNA expression is up-regulated by glucocorticoids and beta2-adrenoceptor agonist in human bronchial epithelial cells and peripheral blood cells (42), and by dexamethasone in ASM cells (3, 33) In A549 cells, transcription is induced by dexamethasone via glucocorticoid receptor occupancy (43)
TNFRSF11B	mRNA expression is up-regulated by dexamethasone in mouse calvarial bones (44)
TSC22D3	Encodes glucocorticoid-induced leucine zipper, known to inhibit the activation of NFκB and mediate most of the anti-inflammatory effects of glucocorticoids (45) mRNA expression is up-regulated by inhaled budesonide in human airways *in vivo* (31, 32) and *in vitro* (3, 33, 34, 45), while down-regulated by inflammatory cytokines (45)
VCAM1	Encodes vascular cell adhesion molecule, a type of immunomodulatory protein expressed by human ASM cells (46) mRNA expression is down-regulated by glucocorticoids in bronchial epithelial cells (47) and ASM cells (3)

Table E3. Differentially Expressed Genes in Fatal Asthma vs. Non-Asthma Donor-Derived ASM Exposed to Budesonide. Two differentially expressed genes had q-value <0.05.

Gene Symbol	Gene Name	Ensembl ID	Genomic Coordinates	Log_2 Fold Change	Q-value	Mean Normalized Counts	
						Asthma	Non-Asthma
CCK	cholecystokinin	ENSG00000187094	chr3:42257825-42266207	-1.74	2.21E-02	9.06	25.69
PMEL	premelanosome protein	ENSG00000185664	chr12:55954105-55973317	-2.25	2.21E-02	24.57	117.41

Table E4. Gene Set Enrichment Analysis of KEGG and Reactome Pathways. Significantly enriched pathways in non-asthma and fatal asthma donor-derived ASM (q-value <0.05) were selected and ordered by q-value. Pathway annotations were downloaded from MSigDB collections. Genes in the leading edge were sorted based on decreasing z-score from DESeq2 results.

Non-Asthma Donor-Derived ASM				
Pathway	**Size**	**Q-value**	**NES**	**Genes in Leading Edge**
KEGG: cytokine-cytokine receptor interaction	163	0.018	-1.97	*IFNAR2*, *IL6ST*, *CCL28*, *EGFR*, *CCR3*, *KIT*, *TNFRSF25*, *CCL13*, *IFNGR2*, *TGFBR1*, *VEGFC*, *CX3CL1*, *CCL8*, *TNFSF9*, *XCR1*, *INHBE*, *IL7R*, *IL13RA1*, *TNFRSF10C*, *BMPR1A*, *CCL7*, *TSLP*, *TNFRSF12A*, *VEGFA*, *TGFB2*, *TNFRSF10B*, *CCR1*, *IL1R1*, *IL12A*, *TNFRSF10D*, *IL6*, *HGF*, *OSMR*, *CCL11*, *GDF5*, *TGFB3*, *LIF*, *IL20RA*, *FIGF*, *CCL2*, *IL17RB*, *TNFSF15*, *CXCL16*, *TNFSF13B*, *TNFSF10*, *TNFRSF19*, *EDA2R*, *PDGFC*, *CXCL12*, *TNFRSF11B*
Reactome: chemokine receptors bind chemokines	27	0.018	-2.10	*CCL28*, *CCR3*, *CX3CL1*, *XCR1*, *CCL7*, *CCR1*, *CCL11*, *CCL2*, *CXCL16*, *CXCL12*
Reactome: antigen presentation folding assembly and peptide loading of class I MHC	21	0.018	-1.93	*B2M*, *TAP2*, *CANX*, *SEC23A*, *SEC24D*, *ERAP1*, *PDIA3*, *CALR*, *SEC31A*, *HSPA5*
KEGG: protein export	23	0.018	-1.89	*SEC61G*, *IMMP2L*, *SEC62*, *SEC61A2*, *SRP72*, *SRP9*, *SPCS2*, *SEC63*, *SEC61A1*, *HSPA5*
KEGG: valine leucine and isoleucine degradation	43	0.018	2.06	*ALDH6A1*, *ABAT*, *MCCC1*, *MCCC2*, *ALDH7A1*, *AOX1*, *HADHB*, *EHHADH*, *ALDH2*, *BCAT2*, *HADHA*, *HADH*, *DBT*, *ACADS*, *HSD17B10*, *IVD*, *MUT*, *ACAA2*, *ACAA1*, *ACAT2*, *OXCT1*
Reactome: metabolism of amino acids and derivatives	161	0.018	1.84	*GLUL*, *ALDH6A1*, *AASS*, *CTH*, *SAT1*, *MCCC1*, *CDO1*, *MCCC2*, *PSMC5*, *ALDH7A1*, *ADI1*, *DIO3*, *SHMT1*, *SQRDL*, *DLST*, *ALDH4A1*, *GLUD1*, *BCAT2*, *SRM*, *TST*, *PRODH*, *CNDP2*, *PSMD8*, *ODC1*, *CBS*, *CKB*, *KYNU*, *DBT*, *GCLM*, *GCDH*, *HSD17B10*, *IVD*, *INHBB*, *AHCY*, *GLS2*, *ASL*, *GOT1*, *MAT2B*, *GSTZ1*, *PSMA7*, *AMD1*, *MTR*, *PCSK1*, *FTCD*, *INHBA*, *SUOX*, *IYD*, *INHA*, *PSMB4*, *PSMB7*, *ALDH9A1*, *OGDH*, *ACAT1*, *PSMD7*, *PSMC4*, *PAOX*, *SLC25A10*, *AZIN1*
KEGG: focal adhesion	184	0.018	1.73	*PIK3R1*, *ITGA10*, *LAMA2*, *CAV1*, *COL4A1*, *LAMC1*, *COL4A4*, *COL5A3*, *COL5A2*, *PAK1*, *LAMB1*, *PPP1CB*, *CAV2*, *BIRC3*, *KDR*, *COL11A1*, *LAMC2*, *CCND3*, *TLN2*, *RAF1*, *DIAPH1*, *LAMA3*, *RAC3*, *COL4A2*, *PDGFD*, *VCL*, *IGF1R*, *PPP1R12A*, *ITGA6*, *SHC3*, *MYL12B*, *PIK3CA*, *ITGA1*, *PXN*, *COMP*, *SOS2*, *ITGA5*, *PDGFA*, *LAMB4*, *FYN*, *ROCK2*, *COL6A3*, *RAC1*, *ITGB1*
KEGG: calcium signaling pathway	148	0.038	-1.60	*CACNA1G*, *PTAFR*, *GRPR*, *PDGFRA*, *VDAC1*, *ADCY9*, *GNAQ*, *TBXA2R*, *PLN*, *PHKB*, *PTGFR*, *RYR2*, *EGFR*, *PRKACB*, *CALM2*, *ADRA1D*, *CAMK2G*, *HTR2A*, *BDKRB2*, *EDNRA*, *PPP3CB*, *CAMK2A*, *DRD1*, *PLCB1*, *ADCY4*, *ADORA2B*, *ATP2B4*, *PHKG1*, *CACNA1C*, *HRH1*, *GRIN2A*, *HTR2B*, *P2RX7*, *ITPKB*, *BDKRB1*, *CHRM2*, *GNA14*, *ADCY8*, *CCKAR*, *SLC8A1*, *ADCY1*
Reactome: metabolism of lipids and lipoproteins	404	0.040	1.45	*PIK3R1*, *HSD11B1*, *CAV1*, *TXNRD1*, *PPARG*, *ARNTL*, *ABHD5*, *DHCR24*, *ACSL1*, *CYP19A1*, *LGMN*, *CERS6*, *PPP1CB*, *PCYT2*, *FABP4*, *PNPLA8*, *GPD1L*, *FAR2*, *MTMR14*, *ELOVL1*, *SLC44A1*, *TM7SF2*, *HADHB*, *ACACB*, *PLIN2*, *ETNK2*, *GPD2*, *CRLS1*, *SUMF2*, *INPP5K*, *HSD17B4*, *PEMT*, *HADHA*, *CDIPT*, *ABCG1*, *MED27*, *GM2A*, *NPAS2*, *PHYH*, *NCOA3*, *VAPA*, *SRD5A3*, *PRKAG2*, *HADH*, *CYP7B1*, *PNPLA2*, *PIK3CA*, *PPARA*, *ABCA1*, *BDH1*, *ACADS*, *CPT1A*, *TNFRSF21*, *MED18*, *SLC25A1*, *NFYB*, *ARSK*, *CERS5*, *CYP1A1*, *CROT*, *MUT*, *GPCPD1*, *TEAD1*, *ACAA1*, *SLC44A2*, *OXCT1*, *TEAD3*, *ANKRD1*, *PLD2*, *BMP1*, *CYP2R1*, *PCYT1A*, *NCOA2*, *MED12*, *CDK19*, *PPP1CA*, *MED8*, *TAZ*, *MTTP*, *GBA2*, *PPM1L*, *SIN3A*, *ELOVL6*, *ACER3*, *HSD17B7*, *PEX11A*, *PIK3C2B*, *HMGCL*, *ECHS1*, *CHPT1*, *MED10*, *ELOVL5*, *ARF3*, *ESRRA*, *CHKA*, *AGPAT1*, *ABCD1*, *EP300*, *CRAT*, *MED30*, *ECI1*, *INPP5E*, *MTM1*, *ACOX1*, *HSD17B3*, *ARF1*, *ACAT1*, *GALC*

Reactome: transport of glucose and other sugars bile salts and organic acids metal ions and amine compounds	63	0.040	-1.70	*SLC41A1*, *SLC41A2*, *SLC39A8*, *SLC16A1*, *SLC2A3*, *CP*, *SLC39A7*, *SLC2A12*, *SLC44A3*, *SLC47A1*, *SLC5A3*, *SLC39A10*, *SLC14A1*, *SLC2A5*, *SLC6A9*, *SLC6A6*
Reactome: cytosolic tRNA amino acylation	24	0.040	-1.88	*LARS*, *CARS*, *DARS*, *EPRS*, *NARS*, *AIMP1*, *GARS*, *PPA1*, *TARS*, *YARS*, *AARS*, *IARS*, *WARS*
Reactome: nitric oxide stimulates guanylate cyclase	24	0.046	-1.84	*PDE9A*, *GUCY1B3*, *MRVI1*, *PRKG1*, *GUCY1A3*, *PRKG2*, *PDE5A*, *GUCY1A2*
KEGG: gap junction	79	0.046	-1.71	*TUBB4B*, *PDGFRA*, *ADCY9*, *GNAQ*, *GJA1*, *TUBA1C*, *TUBB3*, *EGFR*, *SOS1*, *PRKACB*, *KRAS*, *HTR2A*, *TUBA1B*, *GUCY1B3*, *DRD1*, *PLCB1*, *ADCY4*, *PRKG1*, *HTR2B*, *TUBB*, *GUCY1A3*, *ADCY8*, *PRKG2*, *TUBA1A*, *TUBA4A*, *GUCY1A2*, *PDGFC*, *ADCY1*
KEGG: small cell lung cancer	81	0.046	1.72	*PIK3R1*, *LAMA2*, *NFKBIA*, *COL4A1*, *LAMC1*, *COL4A4*, *LAMB1*, *BIRC3*, *LAMC2*, *LAMA3*, *COL4A2*, *SKP2*, *MYC*, *ITGA6*, *E2F1*, *CASP9*, *PIK3CA*, *TRAF1*, *LAMB4*, *TRAF6*, *ITGB1*, *RB1*, *BCL2L1*, *APAF1*, *CCNE2*
Fatal Asthma Donor-Derived ASM				
Pathway	**Size**	**Q-value**	**NES**	**Genes in Leading Edge**
Reactome: signaling by GPCR	392	0.010	-1.56	*ADRA1D*, *GPR132*, *PIK3R3*, *CDK1*, *PPP3CB*, *PTGFR*, *CD55*, *GNG12*, *CX3CL1*, *SOS1*, *MAPK1*, *DGKB*, *ADM2*, *MMP3*, *MAP2K1*, *F2RL1*, *PRKACB*, *KRAS*, *PREX1*, *CREB1*, *PDE8B*, *PDE10A*, *PDE4C*, *ARHGEF2*, *PTGIR*, *EGFR*, *TRIO*, *PDE4B*, *MCHR1*, *GRPR*, *P2RY1*, *FZD7*, *RAMP3*, *CCR3*, *ADCY7*, *CALM2*, *SMO*, *CCL7*, *PLCB1*, *ADCY4*, *XCR1*, *PTGER4*, *CXCL16*, *DRD1*, *P2RY12*, *CCL2*, *CCL28*, *S1PR3*, *KALRN*, *ADORA2B*, *RPS6KA3*, *GNA14*, *AGT*, *CCL11*, *PLA2G4A*, *RGS6*, *GNG2*, *EDN1*, *RXFP1*, *PDE7B*, *EDNRA*, *GPR68*, *RGS4*, *ARHGEF3*, *ADORA1*, *BDKRB2*, *HRH1*, *PPP1R1B*, *CHRM2*, *BDKRB1*, *CCR1*, *CXCL12*, *PRKCE*, *RASGRF2*, *ADCY8*, *CCKAR*, *SSTR1*, *HTR2B*, *ADRA2A*, *ADCY1*, *WNT2*
Reactome: transmembrane transport of small molecules	312	0.010	-1.53	*SLC7A1*, *NUP35*, *ATP2C1*, *SLC28A3*, *SLC24A4*, *SLC29A4*, *SLC39A8*, *SLC2A1*, *ABCA8*, *ADCY9*, *TFRC*, *SLC20A2*, *ATP2A2*, *ABCB5*, *SLC38A1*, *SLC44A3*, *SLC38A4*, *GNG12*, *PEX3*, *CP*, *SLC30A1*, *SLC46A1*, *SLC41A1*, *SLC35B4*, *SLC16A7*, *ATP9A*, *PRKACB*, *SLC2A12*, *ATP6V1B2*, *SLC12A2*, *SLC16A1*, *ATP1A2*, *NUP205*, *SLC1A7*, *ATP1B3*, *ATP6V1C1*, *HK2*, *SLC7A5*, *SLC41A2*, *SLC2A3*, *ADCY7*, *GLRB*, *SLC1A3*, *ATP8B4*, *ADCY4*, *AQP1*, *SLC6A9*, *AQP7*, *SLC47A1*, *SLCO3A1*, *GNG2*, *SLC1A2*, *ATP7A*, *ABCA10*, *ATP6V1A*, *SLC9A7*, *ABCC6*, *SLC38A5*, *SLC14A1*, *SLC4A4*, *ADCY8*, *STEAP3*, *SLC2A5*, *SLC39A10*, *SLC5A3*, *SLC8A1*, *ATP8B1*, *AQP3*, *SLC6A6*, *SLC4A7*, *ADCY1*
KEGG: cytokine-cytokine receptor interaction	167	0.010	-2.20	*CX3CL1*, *TNFRSF25*, *IL13RA1*, *BMPR2*, *TNFRSF12A*, *GDF5*, *IL7R*, *TNFSF9*, *EGFR*, *VEGFC*, *IL20RA*, *CCL13*, *CCR3*, *INHBE*, *TNFRSF10C*, *CCL7*, *IL1R1*, *TNFRSF10D*, *VEGFA*, *BMPR1A*, *XCR1*, *CXCL16*, *OSMR*, *CCL8*, *CCL2*, *CCL28*, *IL12A*, *TSLP*, *KIT*, *TNFSF15*, *CCL11*, *IL17RB*, *TNFRSF10B*, *HGF*, *CCR1*, *TGFB3*, *CXCL12*, *TGFBR1*, *PDGFC*, *TNFSF10*, *TGFB2*, *LIF*, *IL6*, *TNFSF13B*, *EDA2R*, *FIGF*, *TNFRSF19*, *TNFRSF11B*
Reactome: G alpha (i) signaling events	111	0.010	-1.78	*MCHR1*, *CCR3*, *ADCY7*, *ADCY4*, *CXCL16*, *P2RY12*, *CCL28*, *S1PR3*, *AGT*, *RGS6*, *GNG2*, *RGS4*, *ADORA1*, *BDKRB2*, *CHRM2*, *BDKRB1*, *CCR1*, *CXCL12*, *ADCY8*, *SSTR1*, *ADRA2A*, *ADCY1*
Reactome: phospholipase C mediated cascade	45	0.010	-1.96	*ADCY9*, *FGF1*, *KL*, *FGFR4*, *PRKACB*, *CREB1*, *FGF2*, *ADCY7*, *CALM2*, *ADCY4*, *FGF7*, *FGFR2*, *PRKCE*, *ADCY8*, *FGF5*, *ADCY1*
Reactome: chemokine receptors bind chemokines	27	0.010	-2.05	*CX3CL1*, *CCR3*, *CCL7*, *XCR1*, *CXCL16*, *CCL2*, *CCL28*, *CCL11*, *CCR1*, *CXCL12*
KEGG: ribosome	86	0.010	2.11	*RPL26L1*, *RPS7*, *RPL17*, *RPL14*, *RPLP0*, *RPL10*, *RPL38*, *RPL29*, *RPS29*, *RPL36*, *UBA52*, *RPL28*,

				RPL11, *RPL4*, *RPL35*, *RPS18*, *RPS10*, *RPS3*, *RPL27*, *RPS9*, *RPS28*, *RPL12*, *RPL18*, *RPL19*, *RPL8*, *RPLP2*, *RPL37*, *RPL18A*, *RPL7A*, *RPS8*, *RPLP1*, *RPS27*, *RPS11*, *RPS6*, *RPS2*, *RPL37A*, *FAU*, *RPL10A*, *RPL23A*, *RPS5*, *RPSA*, *RPS20*, *RPS26*, *RPL32*, *RPL41*, *RPL13*, *RPS19*, *RPS25*, *RPS16*, *RPS15*, *RPL24*, *RPL13A*, *RPL3*, *RPL6*, *RPS4X*, *RPL23*, *RPL15*, *RPL30*
Reactome: peptide chain elongation	102	0.010	2.14	*RPL26L1*, *RPS7*, *RPL17*, *RPL14*, *RPLP0*, *RPL10*, *RPL38*, *RPL29*, *RPS29*, *RPL36*, *UBA52*, *RPL28*, *RPL11*, *RPL4*, *RPL35*, *RPS18*, *RPS10*, *RPS3*, *RPS28P7*, *RPL27*, *RPS9*, *RPS28*, *RPL12*, *RPL18*, *FAM153A*, *EEF2*, *RPL19*, *RPL8*, *RPLP2*, *RPL37*, *RPL18A*, *RPS14*, *RPL7A*, *RPS8*, *RPLP1*, *RPS27*, *RPS11*, *RPS6*, *RPS2*, *RPL37A*, *FAU*, *RPL10A*, *RPL23A*, *RPS5*, *RPSA*, *RPS20*, *RPS26*, *RPL32*, *RPL41*, *RPL13*, *RPS19*, *RPS25*, *RPS16*, *RPS15*, *RPL24*, *RPL13A*, *RPL3*, *RPL6*, *RPS19P3*, *RPS4X*, *RPL23*, *RPL15*, *RPL30*, *RPL23AP42*, *RPL23AP2*, *RPL31*, *RPL21P134*, *RPL36A*, *RPS24*
Reactome: metabolism of amino acids and derivatives	160	0.010	1.79	*GLUL*, *ALDH6A1*, *AASS*, *DIO3*, *SAT1*, *MCCC1*, *BCAT2*, *CTH*, *ADI1*, *CBS*, *PSMC5*, *TST*, *CDO1*, *SHMT1*, *ALDH7A1*, *SRM*, *ALDH4A1*, *AHCY*, *MCCC2*, *CKB*, *ODC1*, *SQRDL*, *PSMD8*, *INHBB*, *DLST*, *KYNU*, *ASL*, *GLS2*, *CNDP2*, *GCDH*, *DBT*, *GLUD1*, *INHA*, *MAT2B*, *GSTZ1*, *HSD17B10*, *IVD*, *CSAD*, *SUOX*, *FTCD*, *PSMA7*, *PSMC4*, *PRODH*, *PSMD7*, *PAOX*, *PCSK1*, *AFMID*, *PSMC3*, *MRI1*, *GCLM*, *PSMB4*, *PSMD4*, *GRHPR*, *PSMB10*, *SLC25A21*, *PCBD1*, *PSMB7*, *ALDH9A1*, *GOT1*
KEGG: gap junction	79	0.019	-1.78	*PDGFRA*, *TUBB4B*, *TUBA1C*, *HTR2A*, *GJA1*, *ADCY9*, *CDK1*, *SOS1*, *MAPK1*, *MAP2K1*, *PRKACB*, *KRAS*, *EGFR*, *ADCY7*, *TUBA1B*, *PLCB1*, *GUCY1B3*, *ADCY4*, *DRD1*, *PRKG1*, *TUBB*, *TUBA4A*, *PRKG2*, *GUCY1A2*, *TUBA1A*, *PDGFC*, *ADCY8*, *HTR2B*, *GUCY1A3*, *ADCY1*
Reactome: nitric oxide stimulates guanylate cyclase	24	0.019	-1.90	*PDE10A*, *PDE9A*, *MRVI1*, *GUCY1B3*, *PRKG1*, *PRKG2*, *GUCY1A2*, *PDE5A*, *GUCY1A3*
Reactome: GPCR ligand binding	203	0.023	-1.60	*ADRA1D*, *GPR132*, *PTGFR*, *CD55*, *GNG12*, *CX3CL1*, *ADM2*, *F2RL1*, *PTGIR*, *MCHR1*, *GRPR*, *P2RY1*, *FZD7*, *RAMP3*, *CCR3*, *SMO*, *CCL7*, *XCR1*, *PTGER4*, *CXCL16*, *DRD1*, *P2RY12*, *CCL2*, *CCL28*, *S1PR3*, *ADORA2B*, *AGT*, *CCL11*, *GNG2*, *EDN1*, *RXFP1*, *EDNRA*, *GPR68*, *ADORA1*, *BDKRB2*, *HRH1*, *CHRM2*, *BDKRB1*, *CCR1*, *CXCL12*, *CCKAR*, *SSTR1*, *HTR2B*, *ADRA2A*, *WNT2*
KEGG: TGF beta signaling pathway	79	0.023	-1.73	*MAPK1*, *RPS6KB1*, *BMPR2*, *GDF5*, *ID4*, *SMAD1*, *ZFYVE16*, *NOG*, *INHBE*, *E2F5*, *BMPR1A*, *ACVR1C*, *ZFYVE9*, *GDF6*, *SMAD5*, *ID2*, *SMAD6*, *TGFB3*, *THBS2*, *TGFBR1*, *TGFB2*, *FST*, *SMAD3*
Reactome: voltage gated potassium channels	31	0.023	-1.81	*KCND3*, *KCNH2*, *KCNQ5*, *KCNA5*, *KCNAB1*, *KCNH5*, *KCNS1*, *KCND2*, *KCNA3*, *KCNB2*
Reactome: neurotransmitter receptor binding and downstream transmission in the postsynaptic cell	109	0.026	-1.68	*DLG1*, *RPS6KA6*, *ADCY9*, *KCNJ15*, *CHRNA1*, *GNG12*, *MAPK1*, *PRKACB*, *NCALD*, *CREB1*, *NSF*, *AKAP5*, *MDM2*, *ADCY7*, *CALM2*, *CAMK2A*, *PLCB1*, *ADCY4*, *GRIP2*, *KCNJ2*, *RPS6KA3*, *GNG2*, *DLG3*, *GRIN2A*, *RASGRF2*, *ADCY8*, *GRIK4*, *ADCY1*
Reactome: metabolism of lipids and lipoproteins	408	0.030	1.38	*PPARG*, *ELOVL1*, *DHCR24*, *CAV1*, *PIK3R1*, *ACSL1*, *ARNTL*, *HSD11B1*, *CYP19A1*, *TXNRD1*, *PPP1CB*, *FABP4*, *NPAS2*, *PCYT2*, *ABHD5*, *LGMN*, *ETNK2*, *PLIN2*, *PEMT*, *MTMR14*, *GPD2*, *LPIN3*, *TM7SF2*, *HADHB*, *CYP7B1*, *HADHA*, *CERS6*, *TNFRSF21*, *NCOA3*, *CPT1A*, *CDIPT*, *PNPLA2*, *BDH1*, *ABCA1*, *ACADS*, *SUMF2*, *MED27*, *PNPLA8*, *INPP5K*, *ACACB*, *OXCT1*, *ECI1*, *CRLS1*, *HMGCL*, *PRKAG2*, *HSD17B4*, *LIPE*, *FAR2*, *CROT*, *ELOVL6*, *SLC25A1*, *CYP1A1*, *AGPAT2*, *PHYH*, *TEAD3*, *ANKRD1*, *SLC44A2*, *GPD1L*, *ABCG1*, *GM2A*, *PIK3CA*, *MED25*, *VAPA*, *PPP1CA*, *TAZ*, *ACAA1*, *SLC44A1*, *ARSK*, *BMP1*, *MED18*, *ABCD1*, *SRD5A3*, *ESRRA*, *PPARA*, *PLD2*, *NFYB*, *MED8*, *ANGPTL4*, *CYP39A1*, *GBA2*, *CRAT*, *PIK3CD*, *CYP2R1*, *ARF1*, *GPCPD1*, *CERS5*, *AGPAT1*
Reactome: sulfur amino acid metabolism	23	0.040	1.86	*CTH*, *ADI1*, *CBS*, *TST*, *CDO1*, *AHCY*, *SQRDL*, *CNDP2*, *MAT2B*, *CSAD*, *SUOX*, *MRI1*, *GCLM*, *GOT1*

Table E5. Publicly Available Glucocorticoid Response Transcriptomic Datasets Selected for Analysis.

GEO ID	Cell Type	Treatment Description	Control Samples (N)	Treated Samples (N)
*GSE94335	Non-asthma donor-derived ASM	Budesonide (100 nM, 24 hr)	8	8
*GSE94335	Fatal asthma donor-derived ASM	Budesonide (100 nM, 24 hr)	9	9
GSE13168	ASM	Fluticasone (10 nM, 0.5 hr)	3	3
GSE34313	ASM	Dexamethasone (1 μM, 24 hr)	3	3
SRP033351	ASM	Dexamethasone (1 μM, 18 hr)	4	4
GSE1815	Bronchial epithelium	Dexamethasone (100 nM, 8 hr)	4	4
GSE55876	Childhood acute lymphoblastic leukemia cell	Dexamethasone (100 nM, 6 hr)	3	3
GSE55877	Childhood acute lymphoblastic leukemia cell	Dexamethasone (100 nM, 6 hr)	3	3
GSE22152	Childhood acute lymphoblastic leukemia cell	Dexamethasone (100 nM, 6 hr)	6	6
GSE44248	Lymphoblastoid cell line	Dexamethasone (1 μM, 16 hr)	8	8
GSE61880	Macrophage	Dexamethasone (100 nM, 10 hr)	3	3
GSE79077	Macrophage	Dexamethasone (100 nM, 20 hr)	3	3
GSE87000	Macrophage	Dexamethasone (100 nM, 6 hr)	7	7
GSE4917	MCF10A-Myc cell	Dexamethasone (1 μM, 24 hr)	3	3
GSE46448	Osteosarcoma U2OS cell	Dexamethasone (100 nM, 4 hr)	4	4

* Also the current study.

Table E6. Significant genes (N=154) identified by integration of four non-asthma ASM studies of glucocorticoid response (including the non-asthma donor-derived cells reported in Table E2). 56 genes shown in bold were present in all 14 datasets and used for hierarchical clustering. For rank-based method, if all simulated rank products were larger than the observed rank product, the corresponding p-values and FDR were zero.

Gene	Effect-size-based method							Rank-based method		
	Log_2 Fold Change	**SE**	**95% CI lower bound**	**95% CI upper bound**	**df**	**P-value**	**Q-value**	**Rank Product**	**P-value**	**FDR**
GPM6B	2.40	0.06	2.27	2.52	3	9.05E-300	8.69E-296	0.258	0	0
FAM107A	4.81	0.24	4.35	5.28	3	1.77E-91	2.13E-88	0.227	0	0
GPX3	4.11	0.23	3.66	4.56	3	1.18E-71	1.13E-68	0.261	0	0
GLUL	2.92	0.22	2.49	3.34	3	1.06E-41	3.92E-39	0.253	0	0
CCDC69	2.15	0.24	1.68	2.62	3	2.75E-19	2.61E-17	0.127	0	0
MEST	-2.28	0.26	-2.78	-1.78	3	3.55E-19	3.30E-17	0.335	0	0
SAMHD1	3.04	0.35	2.35	3.72	3	2.81E-18	2.49E-16	0.437	0	0
METTL7A	2.32	0.33	1.68	2.96	3	1.14E-12	5.78E-11	0.335	0	0
SORT1	1.81	0.26	1.29	2.32	3	4.57E-12	2.20E-10	0.462	0	0
ZBTB16	5.41	0.78	3.88	6.94	3	4.72E-12	2.26E-10	0.523	0	0
FKBP5	3.61	0.53	2.58	4.64	3	6.93E-12	3.24E-10	0.237	0	0
TSC22D3	3.50	0.56	2.40	4.59	3	3.76E-10	1.36E-08	0.501	0	0
MAOA	2.82	0.49	1.85	3.79	3	1.12E-08	3.02E-07	0.326	0	0
CRISPLD2	2.64	0.47	1.71	3.56	3	2.12E-08	5.45E-07	0.172	0	0
DUSP1	1.99	0.52	0.98	3.00	3	1.09E-04	1.24E-03	0.205	0	0
CACNB2	2.27	0.70	0.89	3.65	3	1.23E-03	9.62E-03	0.118	0	0
FOXO1	2.39	0.09	2.22	2.57	3	4.61E-156	1.48E-152	0.240	1.00E-06	9.37E-04
CITED2	1.93	0.25	1.44	2.42	3	1.30E-14	8.31E-13	0.155	1.00E-06	9.37E-04
SPARCL1	3.40	0.55	2.32	4.49	3	6.85E-10	2.34E-08	0.463	1.00E-06	9.37E-04
WNT2	-2.42	0.43	-3.25	-1.58	3	1.33E-08	3.58E-07	0.520	1.00E-06	9.37E-04
ITGA10	2.84	0.59	1.68	4.01	3	1.68E-06	3.01E-05	0.294	1.00E-06	9.37E-04
KLF15	3.22	1.19	0.89	5.55	3	6.80E-03	3.83E-02	0.304	1.00E-06	9.37E-04
TEX2	1.25	0.14	0.98	1.51	3	5.98E-20	6.14E-18	0.481	2.00E-06	1.57E-03
IMPA2	2.50	0.41	1.71	3.30	3	7.05E-10	2.40E-08	0.290	2.00E-06	1.57E-03
PER1	3.02	0.17	2.68	3.35	2	1.11E-70	1.01E-67	0.675	3.00E-06	2.06E-03
SOX4	-2.02	0.18	-2.37	-1.67	3	2.62E-30	5.31E-28	0.463	3.00E-06	2.06E-03
GDF15	-2.07	0.41	-2.87	-1.27	3	4.02E-07	8.08E-06	0.249	3.00E-06	2.06E-03
IL16	1.47	0.29	0.89	2.04	3	5.24E-07	1.03E-05	0.288	3.00E-06	2.06E-03
TNFAIP6	-2.96	0.85	-4.62	-1.31	3	4.51E-04	4.13E-03	0.492	3.00E-06	2.06E-03
ADAM12	-1.71	0.15	-1.99	-1.42	3	2.68E-31	5.60E-29	0.026	4.00E-06	2.44E-03
TNFRSF11B	-2.29	0.83	-3.92	-0.67	3	5.73E-03	3.35E-02	0.493	4.00E-06	2.44E-03
TRNP1	1.76	0.19	1.38	2.14	2	1.35E-19	1.32E-17	0.760	5.00E-06	2.91E-03
TGFBR2	1.20	0.11	0.98	1.42	3	1.62E-26	2.61E-24	0.482	6.00E-06	3.39E-03
PIK3R1	1.67	0.17	1.34	2.00	3	4.75E-23	6.08E-21	0.390	7.00E-06	3.79E-03
ANXA4	1.12	0.18	0.76	1.48	3	1.25E-09	4.13E-08	0.060	7.00E-06	3.79E-03
KCNK6	2.42	0.18	2.07	2.76	2	7.32E-43	2.93E-40	0.546	8.00E-06	4.10E-03

CXCL12	-2.13	0.38	-2.87	-1.39	3	1.68E-08	4.40E-07	0.181	8.00E-06	4.10E-03
STOM	1.29	0.05	1.18	1.40	3	2.48E-123	6.81E-120	0.471	9.00E-06	4.32E-03
PRSS35	-2.24	0.23	-2.70	-1.79	2	1.05E-21	1.23E-19	0.577	9.00E-06	4.32E-03
MT1E	1.95	0.27	1.42	2.47	3	3.12E-13	1.69E-11	0.346	9.00E-06	4.32E-03
RASL11A	2.55	0.45	1.66	3.44	2	1.99E-08	5.14E-07	0.744	1.10E-05	5.03E-03
ADAMTS1	1.94	0.19	1.56	2.31	3	6.19E-24	8.26E-22	0.028	1.20E-05	5.18E-03
SMARCD2	1.26	0.15	0.97	1.55	3	1.59E-17	1.32E-15	0.458	1.20E-05	5.18E-03
LAMA2	1.71	0.35	1.01	2.40	3	1.33E-06	2.43E-05	0.309	1.20E-05	5.18E-03
VCAM1	-2.67	0.69	-4.01	-1.32	3	1.00E-04	1.16E-03	0.514	1.20E-05	5.18E-03
MT2A	1.55	0.56	0.46	2.65	3	5.44E-03	3.22E-02	0.346	1.40E-05	5.72E-03
TCEAL1	1.09	0.10	0.89	1.28	3	7.63E-28	1.31E-25	0.479	1.80E-05	7.13E-03
ARMC8	1.11	0.12	0.88	1.35	3	4.52E-21	4.96E-19	0.074	2.00E-05	7.69E-03
TJP2	1.92	0.15	1.63	2.22	3	4.37E-38	1.27E-35	0.486	2.10E-05	7.84E-03
FGD4	2.05	0.33	1.40	2.70	2	5.40E-10	1.89E-08	0.747	2.10E-05	7.84E-03
IRS2	1.92	0.17	1.59	2.25	3	6.77E-30	1.31E-27	0.293	2.20E-05	7.98E-03
SCD	-1.64	0.51	-2.64	-0.63	3	1.41E-03	1.07E-02	0.438	2.30E-05	8.19E-03
NOV	-2.11	0.45	-2.98	-1.23	3	2.30E-06	3.98E-05	0.362	2.60E-05	9.08E-03
FBN2	2.16	0.26	1.64	2.68	3	3.27E-16	2.41E-14	0.231	2.70E-05	9.27E-03
PPP1R14A	3.07	0.10	2.88	3.26	2	1.40E-226	8.98E-223	0.703	2.80E-05	9.28E-03
ZHX3	1.03	0.08	0.87	1.20	3	5.24E-34	1.29E-31	0.526	2.80E-05	9.28E-03
WARS	-1.06	0.19	-1.44	-0.68	3	5.19E-08	1.25E-06	0.517	2.80E-05	9.28E-03
KLF9	1.77	0.19	1.40	2.13	3	2.28E-21	2.60E-19	0.304	2.90E-05	9.29E-03
FST	-1.95	0.09	-2.12	-1.78	3	2.39E-112	4.59E-109	0.241	3.30E-05	9.76E-03
PDE5A	-1.15	0.13	-1.40	-0.90	3	1.97E-19	1.90E-17	0.382	3.20E-05	9.76E-03
CDH6	-1.52	0.26	-2.04	-1.00	3	8.87E-09	2.45E-07	0.140	3.20E-05	9.76E-03
MOB3B	2.47	0.76	0.98	3.96	3	1.13E-03	8.94E-03	0.340	3.20E-05	9.76E-03
SYNJ2	-1.33	0.22	-1.75	-0.91	3	6.67E-10	2.29E-08	0.474	3.10E-05	9.77E-03
ERRFI1	2.00	0.32	1.38	2.62	2	2.85E-10	1.05E-08	0.576	3.50E-05	1.01E-02
SCN7A	1.62	0.36	0.91	2.33	3	7.35E-06	1.14E-04	0.439	3.50E-05	1.01E-02
HSPA2	1.81	0.28	1.26	2.36	3	9.43E-11	3.75E-09	0.280	3.60E-05	1.02E-02
ANGPTL7	3.73	1.26	1.27	6.19	3	2.96E-03	1.98E-02	0.055	3.70E-05	1.03E-02
WIPI1	1.09	0.18	0.72	1.45	3	4.01E-09	1.20E-07	0.520	4.10E-05	1.13E-02
NR3C1	-1.10	0.12	-1.33	-0.87	3	7.17E-21	7.83E-19	0.364	4.40E-05	1.17E-02
GLRX	0.89	0.21	0.49	1.30	3	1.81E-05	2.53E-04	0.253	4.40E-05	1.17E-02
CEBPD	1.43	0.24	0.97	1.90	3	2.06E-09	6.52E-08	0.144	4.80E-05	1.25E-02
BDKRB1	-2.71	0.57	-3.83	-1.59	3	2.08E-06	3.63E-05	0.098	5.00E-05	1.28E-02
ALDH6A1	1.15	0.23	0.71	1.60	3	3.56E-07	7.27E-06	0.048	5.40E-05	1.36E-02
DCXR	1.70	0.43	0.87	2.54	3	6.61E-05	8.00E-04	0.189	5.70E-05	1.40E-02
CCND3	1.03	0.06	0.92	1.14	3	3.66E-74	3.90E-71	0.130	5.90E-05	1.44E-02
KCTD12	-1.75	0.39	-2.50	-0.99	3	5.87E-06	9.31E-05	0.300	6.00E-05	1.44E-02
SLC16A12	3.02	0.27	2.50	3.55	2	8.17E-30	1.54E-27	0.866	6.10E-05	1.45E-02
SLC7A8	1.66	0.42	0.83	2.48	3	8.69E-05	1.01E-03	0.456	6.30E-05	1.48E-02
TM2D1	0.51	0.06	0.40	0.62	3	1.74E-19	1.69E-17	0.487	6.80E-05	1.55E-02

MT1X	2.54	0.53	1.50	3.58	3	1.74E-06	3.09E-05	0.346	6.70E-05	1.55E-02
HIF3A	1.84	0.61	0.65	3.04	3	2.52E-03	1.74E-02	0.272	7.00E-05	1.56E-02
NFKBIA	1.32	0.13	1.07	1.58	3	4.64E-24	6.28E-22	0.358	7.10E-05	1.57E-02
SLC7A14	-2.26	0.25	-2.74	-1.77	2	1.19E-19	1.18E-17	0.780	7.60E-05	1.64E-02
ABLIM1	0.72	0.21	0.32	1.13	3	5.02E-04	4.52E-03	0.014	7.70E-05	1.64E-02
PHC2	1.33	0.10	1.14	1.53	3	7.30E-41	2.42E-38	0.387	8.00E-05	1.68E-02
NID1	1.35	0.47	0.44	2.27	3	3.69E-03	2.36E-02	0.358	8.00E-05	1.68E-02
IER3	-2.33	0.57	-3.45	-1.21	3	4.67E-05	5.91E-04	0.283	8.20E-05	1.69E-02
APBB2	1.00	0.10	0.80	1.21	3	9.59E-22	1.14E-19	0.062	8.50E-05	1.74E-02
CPA4	-1.88	0.54	-2.94	-0.81	3	5.62E-04	4.97E-03	0.170	9.30E-05	1.88E-02
HSD11B1	2.45	0.17	2.11	2.79	3	1.86E-45	8.50E-43	0.279	1.03E-04	2.04E-02
PER3	-1.95	0.23	-2.41	-1.49	3	7.16E-17	5.55E-15	0.385	1.03E-04	2.04E-02
SEMA3A	-1.27	0.30	-1.85	-0.70	3	1.59E-05	2.26E-04	0.442	1.03E-04	2.04E-02
ADRA1B	2.63	0.99	0.69	4.57	3	7.77E-03	4.26E-02	0.036	1.08E-04	2.05E-02
HPS5	1.25	0.15	0.97	1.54	3	7.02E-18	6.05E-16	0.278	1.07E-04	2.06E-02
EXT1	-1.06	0.15	-1.37	-0.76	3	6.98E-12	3.26E-10	0.224	1.16E-04	2.19E-02
PDK4	2.82	0.38	2.07	3.58	3	2.18E-13	1.21E-11	0.383	1.20E-04	2.24E-02
RTKN2	-2.37	0.53	-3.42	-1.32	2	9.26E-06	1.40E-04	0.726	1.22E-04	2.25E-02
GPRC5B	1.68	0.14	1.40	1.96	3	2.83E-32	6.49E-30	0.260	1.49E-04	2.60E-02
SUN2	1.54	0.32	0.91	2.16	3	1.34E-06	2.44E-05	0.473	1.48E-04	2.61E-02
ASB13	1.17	0.26	0.66	1.69	3	8.13E-06	1.24E-04	0.078	1.47E-04	2.62E-02
CELF2	1.18	0.12	0.95	1.42	3	2.33E-23	3.07E-21	0.145	1.45E-04	2.62E-02
GADD45B	1.05	0.15	0.77	1.34	3	9.07E-13	4.61E-11	0.245	1.52E-04	2.63E-02
SYNPO2	1.50	0.15	1.21	1.79	2	1.20E-23	1.59E-21	0.699	1.62E-04	2.78E-02
BHLHE40	-1.81	0.62	-3.02	-0.59	3	3.62E-03	2.33E-02	0.099	1.69E-04	2.87E-02
VEGFA	-1.49	0.40	-2.28	-0.71	3	2.04E-04	2.11E-03	0.515	1.72E-04	2.87E-02
CRABP2	-1.63	0.16	-1.95	-1.32	3	3.05E-24	4.19E-22	0.171	1.76E-04	2.88E-02
NEXN	1.39	0.37	0.67	2.12	2	1.50E-04	1.64E-03	0.740	1.76E-04	2.88E-02
RAI2	1.32	0.23	0.88	1.76	3	4.72E-09	1.39E-07	0.418	1.79E-04	2.88E-02
STARD7	0.94	0.18	0.58	1.29	3	2.86E-07	5.96E-06	0.469	1.75E-04	2.90E-02
INPP5A	1.11	0.10	0.91	1.32	3	2.28E-26	3.60E-24	0.291	1.88E-04	2.94E-02
GGT5	1.95	0.58	0.80	3.09	3	8.68E-04	7.19E-03	0.250	1.88E-04	2.94E-02
TRPC6	1.30	0.44	0.44	2.17	3	3.09E-03	2.05E-02	0.501	1.88E-04	2.94E-02
MARCKSL1	-1.15	0.23	-1.61	-0.69	3	7.56E-07	1.45E-05	0.329	1.93E-04	2.97E-02
RSPO1	1.65	0.61	0.45	2.84	2	6.80E-03	3.83E-02	0.725	1.87E-04	2.97E-02
IGFBP2	2.60	0.34	1.94	3.27	3	1.65E-14	1.05E-12	0.286	1.99E-04	3.04E-02
CFLAR	0.95	0.16	0.63	1.26	3	3.43E-09	1.04E-07	0.149	2.01E-04	3.04E-02
FER1L6	-2.84	0.13	-3.09	-2.58	1	4.60E-106	8.04E-103	0.596	2.06E-04	3.07E-02
NR2F2	0.92	0.32	0.29	1.55	3	4.32E-03	2.68E-02	0.364	2.05E-04	3.08E-02
TBX18	-1.79	0.08	-1.95	-1.63	1	5.18E-105	8.30E-102	0.775	2.18E-04	3.22E-02
ABAT	1.26	0.11	1.03	1.48	3	1.35E-28	2.45E-26	0.005	2.22E-04	3.26E-02
AFAP1L1	2.02	0.37	1.30	2.75	2	4.87E-08	1.18E-06	0.550	2.26E-04	3.29E-02
CORO6	2.23	0.59	1.08	3.38	2	1.44E-04	1.58E-03	0.608	2.28E-04	3.29E-02

BDKRB2	-1.97	0.07	-2.11	-1.84	3	5.40E-182	2.59E-178	0.098	2.37E-04	3.40E-02
CPPED1	1.06	0.08	0.89	1.22	3	1.64E-36	4.57E-34	0.170	2.46E-04	3.45E-02
RHOJ	-1.43	0.23	-1.89	-0.98	2	7.21E-10	2.44E-08	0.759	2.44E-04	3.46E-02
TIMP4	2.12	0.65	0.85	3.39	3	1.04E-03	8.37E-03	0.485	2.44E-04	3.46E-02
PGBD5	-1.99	0.17	-2.32	-1.66	3	2.16E-32	5.00E-30	0.386	2.62E-04	3.65E-02
ITGBL1	1.12	0.08	0.96	1.27	3	6.11E-46	2.94E-43	0.295	2.68E-04	3.65E-02
OPN3	1.01	0.36	0.30	1.71	3	5.07E-03	3.04E-02	0.371	2.67E-04	3.66E-02
SRPX	1.05	0.05	0.95	1.15	3	1.18E-97	1.75E-94	0.467	2.66E-04	3.68E-02
TACC1	0.96	0.16	0.64	1.27	3	3.76E-09	1.13E-07	0.475	2.84E-04	3.69E-02
LDB2	-1.38	0.15	-1.68	-1.08	3	2.87E-19	2.72E-17	0.312	2.73E-04	3.69E-02
GXYLT2	-1.52	0.07	-1.66	-1.38	1	2.11E-97	2.90E-94	0.632	2.83E-04	3.70E-02
RPS6KA2	0.75	0.12	0.51	1.00	3	1.56E-09	5.07E-08	0.433	2.79E-04	3.71E-02
CD302	1.52	0.26	1.01	2.03	2	6.59E-09	1.87E-07	0.624	2.79E-04	3.71E-02
TCEAL4	1.29	0.16	0.98	1.60	3	3.20E-16	2.36E-14	0.479	2.77E-04	3.72E-02
ACSS1	1.78	0.66	0.49	3.06	2	6.68E-03	3.78E-02	0.543	2.93E-04	3.78E-02
ULK2	1.20	0.06	1.08	1.31	3	6.76E-94	8.67E-91	0.510	2.96E-04	3.79E-02
ADRA2A	-2.33	0.84	-3.98	-0.67	3	5.84E-03	3.40E-02	0.036	3.01E-04	3.83E-02
DAAM2	1.18	0.20	0.78	1.58	3	8.83E-09	2.44E-07	0.185	3.14E-04	3.97E-02
NT5E	-0.87	0.25	-1.37	-0.38	3	4.84E-04	4.40E-03	0.366	3.24E-04	4.07E-02
PER2	-1.17	0.37	-1.90	-0.44	3	1.80E-03	1.31E-02	0.385	3.35E-04	4.18E-02
OMD	1.79	0.21	1.38	2.20	3	8.50E-18	7.26E-16	0.371	3.47E-04	4.27E-02
JADE1	1.17	0.27	0.65	1.69	3	1.18E-05	1.74E-04	0.295	3.50E-04	4.28E-02
MXRA5	-1.31	0.17	-1.64	-0.97	3	1.71E-14	1.08E-12	0.349	3.46E-04	4.29E-02
C1QTNF1	1.09	0.17	0.75	1.43	3	4.20E-10	1.50E-08	0.112	3.69E-04	4.49E-02
CPM	1.73	0.66	0.43	3.02	3	9.04E-03	4.78E-02	0.170	3.78E-04	4.51E-02
ALCAM	1.18	0.45	0.31	2.05	3	8.11E-03	4.40E-02	0.047	3.74E-04	4.52E-02
RNF144B	2.22	0.49	1.27	3.18	2	5.18E-06	8.30E-05	0.768	3.77E-04	4.53E-02
PLPP3	-0.85	0.30	-1.44	-0.25	3	5.23E-03	3.12E-02	0.395	3.90E-04	4.63E-02
ANPEP	1.16	0.13	0.90	1.42	3	3.13E-18	2.73E-16	0.058	4.11E-04	4.85E-02
AQP3	-1.85	0.50	-2.84	-0.86	3	2.37E-04	2.39E-03	0.067	4.18E-04	4.87E-02
GALNT15	1.55	0.26	1.03	2.06	2	4.15E-09	1.24E-07	0.609	4.17E-04	4.89E-02
SLC4A7	-1.19	0.25	-1.67	-0.71	3	1.21E-06	2.25E-05	0.455	4.26E-04	4.93E-02

CI, confidence interval; df, degree of freedom; FDR, false discovery rate; SE, standard error.

Table E7. Significant genes (N = 8) identified by integration of three studies of childhood acute lymphoblastic leukemia. For rank-based method, if all simulated rank products are larger than the observed rank product, the corresponding p-values and FDR will be zero.

Gene	Effect-size-based method							Rank-based method		
	Log_2 Fold Change	SE	95% CI lower bound	95% CI upper bound	df	P-value	Q-value	Rank Product	P-value	FDR
FKBP5	1.86	0.08	1.70	2.03	2	9.64E-112	8.73E-108	0.311	0	0
TNFSF8	2.71	0.35	2.03	3.39	2	6.09E-15	1.58E-12	0.720	1.00E-06	9.05E-03
NR3C1	1.54	0.06	1.43	1.65	2	1.57E-158	2.84E-154	0.478	2.00E-06	1.03E-02
TSC22D3	3.84	0.65	2.56	5.12	2	3.78E-09	3.76E-07	0.739	2.00E-06	1.03E-02
RUNX2	3.16	0.20	2.76	3.56	2	9.03E-54	1.63E-50	0.589	4.00E-06	1.45E-02
NFIL3	1.70	0.10	1.50	1.89	2	6.94E-64	1.57E-60	0.470	5.00E-06	1.51E-02
SMAP2	1.23	0.06	1.11	1.34	2	5.91E-102	3.57E-98	0.632	1.80E-05	4.07E-02
PFKFB2	2.27	0.42	1.44	3.10	2	7.45E-08	5.35E-06	0.515	1.70E-05	4.40E-02

CI, confidence interval; df, degree of freedom; FDR, false discovery rate; SE, standard error.

Table E8. Significant genes (N = 78) identified by integration of three studies of macrophage. For rank-based method, if all simulated rank products are larger than the observed rank product, the corresponding p-values and FDR will be zero.

Gene	Effect-size-based method							Rank-based method		
	Log_2 Fold Change	SE	95% CI lower bound	95% CI upper bound	df	P-value	Q-value	Rank Product	P-value	FDR
FKBP5	3.29	0.17	2.96	3.62	2	2.25E-83	1.89E-79	0.292	0	0
ITPKC	1.50	0.10	1.30	1.71	2	8.09E-47	2.72E-43	0.365	0	0
TSC22D3	2.94	0.35	2.26	3.63	2	4.31E-17	1.95E-14	0.703	0	0
SESN1	2.33	0.35	1.64	3.01	2	2.74E-11	5.62E-09	0.582	0	0
KIF13B	1.26	0.20	0.87	1.65	2	3.21E-10	5.39E-08	0.374	0	0
ALOX15B	4.15	0.67	2.85	5.46	2	4.51E-10	7.43E-08	0.053	0	0
SAP30	2.42	0.46	1.51	3.32	2	1.67E-07	1.57E-05	0.571	0	0
SIAH2	1.69	0.37	0.97	2.41	2	4.30E-06	2.76E-04	0.589	0	0
IL4I1	-2.21	0.53	-3.25	-1.16	2	3.31E-05	1.55E-03	0.357	0	0
PTEN	1.15	0.18	0.79	1.51	2	3.05E-10	5.22E-08	0.527	2.00E-06	2.92E-03
JDP2	1.48	0.41	0.69	2.28	2	2.63E-04	8.29E-03	0.365	2.00E-06	2.92E-03
ZCCHC6	1.68	0.12	1.44	1.92	2	4.46E-44	1.25E-40	0.764	3.00E-06	3.73E-03
DDIT4	2.79	0.32	2.16	3.41	2	2.21E-18	1.20E-15	0.235	4.00E-06	4.48E-03
GRB10	2.54	0.12	2.30	2.78	1	1.83E-93	3.07E-89	0.325	6.00E-06	6.11E-03
NUDT16	1.78	0.24	1.32	2.25	2	3.31E-14	1.09E-11	0.465	6.00E-06	6.11E-03
TFCP2L1	2.42	0.44	1.56	3.28	1	3.40E-08	3.82E-06	0.661	8.00E-06	7.47E-03
ZBTB16	5.02	0.29	4.46	5.59	1	1.36E-68	5.72E-65	0.758	9.00E-06	7.96E-03
TBC1D16	2.37	0.39	1.60	3.14	1	1.64E-09	2.46E-07	0.650	1.10E-05	9.24E-03
PLEKHA7	2.12	0.21	1.71	2.54	1	2.07E-23	1.83E-20	0.502	1.20E-05	9.60E-03
DAAM2	2.70	0.62	1.49	3.91	1	1.16E-05	6.58E-04	0.230	2.50E-05	1.17E-02
HMGB2	1.52	0.35	0.83	2.21	2	1.49E-05	8.09E-04	0.341	2.40E-05	1.17E-02
VASH1	-1.23	0.39	-1.99	-0.46	2	1.61E-03	3.29E-02	0.732	2.40E-05	1.17E-02
TPST1	2.16	0.50	1.18	3.15	2	1.68E-05	8.81E-04	0.693	2.60E-05	1.18E-02
RAB7B	-1.68	0.28	-2.23	-1.13	2	2.49E-09	3.58E-07	0.535	2.30E-05	1.19E-02
GPR68	-2.00	0.67	-3.30	-0.70	2	2.64E-03	4.65E-02	0.323	2.30E-05	1.19E-02
FMN1	1.76	0.20	1.36	2.16	2	6.10E-18	3.01E-15	0.294	2.70E-05	1.19E-02
NFKBIE	-1.07	0.16	-1.38	-0.76	2	2.11E-11	4.48E-09	0.453	1.90E-05	1.20E-02
PIK3IP1	1.64	0.25	1.16	2.13	2	2.53E-11	5.26E-09	0.497	1.90E-05	1.20E-02
TFPI	1.84	0.26	1.33	2.34	2	7.08E-13	1.86E-10	0.661	1.80E-05	1.21E-02
SLC7A11	-3.48	0.18	-3.84	-3.12	2	1.32E-79	7.42E-76	0.608	1.70E-05	1.22E-02
ATF5	-1.42	0.22	-1.85	-1.00	2	6.73E-11	1.33E-08	0.096	1.60E-05	1.22E-02
PLAU	-1.49	0.15	-1.79	-1.20	2	2.41E-23	2.03E-20	0.501	2.20E-05	1.23E-02
KLF9	1.50	0.33	0.86	2.14	2	4.12E-06	2.68E-04	0.377	2.20E-05	1.23E-02
SIRT5	1.01	0.30	0.43	1.59	2	6.63E-04	1.70E-02	0.590	2.20E-05	1.23E-02
MERTK	1.66	0.29	1.10	2.22	2	6.99E-09	9.32E-07	0.421	2.90E-05	1.25E-02
IRAK3	1.50	0.14	1.23	1.76	2	2.79E-28	4.26E-25	0.361	2.10E-05	1.26E-02
TNFSF13B	-1.35	0.32	-1.98	-0.71	2	3.08E-05	1.45E-03	0.686	3.60E-05	1.48E-02

SLA	1.25	0.11	1.03	1.47	2	4.57E-29	7.68E-26	0.591	4.30E-05	1.66E-02
PDK4	3.08	0.56	1.98	4.18	2	3.62E-08	4.03E-06	0.488	4.30E-05	1.66E-02
HIVEP1	-1.54	0.19	-1.91	-1.16	2	1.19E-15	4.64E-13	0.339	4.60E-05	1.72E-02
ENOX2	0.88	0.16	0.56	1.20	2	7.38E-08	7.51E-06	0.266	4.90E-05	1.79E-02
MDM2	1.32	0.43	0.48	2.15	2	1.97E-03	3.83E-02	0.419	5.80E-05	1.87E-02
FRAT1	1.85	0.39	1.10	2.61	2	1.66E-06	1.19E-04	0.297	5.70E-05	1.88E-02
THBS1	3.82	1.23	1.41	6.22	2	1.90E-03	3.73E-02	0.663	5.60E-05	1.88E-02
SH3PXD2B	2.07	0.60	0.89	3.26	2	6.09E-04	1.58E-02	0.587	5.50E-05	1.91E-02
SMAP2	1.21	0.09	1.02	1.39	2	2.73E-38	6.54E-35	0.611	5.40E-05	1.93E-02
GPR34	1.50	0.49	0.55	2.45	2	2.03E-03	3.89E-02	0.322	6.70E-05	2.08E-02
MTMR11	2.28	0.69	0.93	3.63	1	8.96E-04	2.12E-02	0.438	7.30E-05	2.23E-02
JAKMIP2	-1.77	0.18	-2.11	-1.42	2	8.29E-24	7.74E-21	0.365	7.70E-05	2.29E-02
CPM	1.56	0.48	0.62	2.51	1	1.20E-03	2.64E-02	0.214	7.70E-05	2.29E-02
PHC2	0.91	0.21	0.50	1.32	2	1.20E-05	6.79E-04	0.494	9.30E-05	2.65E-02
ORMDL1	0.81	0.20	0.41	1.21	2	6.33E-05	2.67E-03	0.474	1.17E-04	3.22E-02
CIITA	-2.02	0.54	-3.08	-0.96	2	1.87E-04	6.34E-03	0.196	1.20E-04	3.25E-02
GLUL	1.05	0.20	0.65	1.45	2	2.40E-07	2.13E-05	0.314	1.23E-04	3.28E-02
TSPAN14	0.77	0.11	0.56	0.97	2	3.40E-13	9.74E-11	0.705	1.25E-04	3.28E-02
SGMS2	1.49	0.41	0.69	2.30	1	2.66E-04	8.38E-03	0.585	1.30E-04	3.31E-02
TXNIP	1.33	0.20	0.94	1.72	2	1.45E-11	3.12E-09	0.715	1.40E-04	3.41E-02
ICAM1	-0.99	0.25	-1.48	-0.51	2	6.45E-05	2.72E-03	0.349	1.38E-04	3.41E-02
DUSP1	1.82	0.50	0.83	2.80	2	3.13E-04	9.45E-03	0.253	1.37E-04	3.44E-02
USP53	1.62	0.36	0.91	2.33	1	8.13E-06	4.78E-04	0.729	1.50E-04	3.60E-02
RASSF5	-0.93	0.09	-1.10	-0.76	2	1.05E-26	1.26E-23	0.541	1.57E-04	3.72E-02
TMEM39A	1.03	0.08	0.87	1.19	2	2.54E-36	5.33E-33	0.678	1.74E-04	4.06E-02
GADD45B	1.45	0.38	0.71	2.20	2	1.37E-04	5.04E-03	0.303	1.97E-04	4.19E-02
PMS1	0.98	0.20	0.59	1.38	2	1.23E-06	9.14E-05	0.506	2.01E-04	4.22E-02
PTPRE	-1.25	0.32	-1.87	-0.62	2	9.94E-05	3.81E-03	0.529	1.96E-04	4.22E-02
METTL7A	1.49	0.27	0.95	2.02	2	4.79E-08	5.26E-06	0.422	1.95E-04	4.25E-02
IL13RA1	0.91	0.20	0.51	1.31	2	7.71E-06	4.59E-04	0.355	1.90E-04	4.26E-02
PER1	1.74	0.51	0.75	2.73	1	5.98E-04	1.56E-02	0.909	1.93E-04	4.27E-02
PRKCE	1.77	0.39	1.01	2.53	1	4.96E-06	3.12E-04	0.519	2.18E-04	4.47E-02
SRC	-1.21	0.40	-1.99	-0.43	2	2.32E-03	4.24E-02	0.628	2.24E-04	4.53E-02
S1PR2	-1.06	0.19	-1.44	-0.68	2	5.80E-08	6.12E-06	0.569	2.40E-04	4.63E-02
FES	1.17	0.30	0.58	1.77	2	1.04E-04	3.97E-03	0.289	2.35E-04	4.64E-02
ABLIM3	1.87	0.24	1.40	2.35	1	1.10E-14	3.90E-12	0.011	2.38E-04	4.65E-02
SLAMF1	1.80	0.54	0.74	2.85	2	8.78E-04	2.10E-02	0.591	2.47E-04	4.72E-02
LRRK2	1.24	0.11	1.02	1.45	2	3.64E-29	6.80E-26	0.406	2.51E-04	4.74E-02
MTSS1	1.57	0.37	0.85	2.29	2	1.76E-05	9.18E-04	0.439	2.63E-04	4.80E-02
SLC2A6	-1.48	0.49	-2.44	-0.52	2	2.43E-03	4.38E-02	0.600	2.77E-04	4.95E-02
FBLN5	3.03	0.85	1.37	4.68	2	3.41E-04	1.01E-02	0.285	2.75E-04	4.97E-02

CI, confidence interval; df, degree of freedom; FDR, false discovery rate; SE, standard error.

Table E9. *CEBPD* Expression Changes in Response to Glucocorticoid Exposure in all ASM Studies. Expression levels for RNA-Seq studies GSE94335 and SRP033351 are quantified as normalized counts, and for gene expression microarray studies GSE13168 and GSE34313 are quantified as post-RMA intensity.

Study	Log_2 Fold Change	Q-value	Mean Expression Levels	
			Glucocorticoid	Control
GSE94335 fatal asthma	0.48	3.20E-02	1155.43	821.12
GSE94335 non-asthma	0.93	2.71E-05	1453.71	808.02
SRP033351	1.50	1.90E-38	2900.97	1026.54
GSE13168	1.20	3.95E-01	11.16	9.96
GSE34313	2.01	3.02E-06	15.73	13.72
Non-asthma ASM integration	1.43	6.52E-08	N/A	N/A

Table E10. Summary of *CEBPD* and its Glucocorticoid Responsiveness-Related Role.

Gene Name	Ensembl ID	Genomic Coordinates	Glucocorticoid Responsiveness-Related Evidence
CCAAT/enhancer binding protein (C/EBP), delta	ENSG00000221869	chr8:47736909-47739086	Encodes a family of C/EBP transcription factors regulating a wide variety of processes, including inflammation, cell growth arrest and differentiation (48-50) mRNA expression is up-regulated by glucocorticoids in ASM (3, 31), lung epithelium (43, 51), skeletal muscle (52) and adipocytes and adipocytes (53, 54). In A549 cells, transcription is induced by dexamethasone via glucocorticoid receptor occupancy (43) In lymphoblastoid cell lines, mRNA expression is increased in asthma vs. non-asthma subjects (35)

Figure Legends

Figure E1. RNA-Seq data quality control. A) Estimated read coverage across transcripts for each sample. Position and coverage are normalized by adjusting for transcript lengths and total number of reads per sample. B) Correlation between observed ERCC transcript read counts vs. expected ERCC transcript molecules. Use sample 28788 as an example. C) Gene expression of the housekeeping genes *GABARAP* and *RPL19* by condition status (i.e. budesonide vs. control in fatal asthma-derived and non-asthma-derived ASM). RNA-Seq results showed high levels of expression for each gene that did not significantly differ by condition status. BUD, budesonide.

Figure E2. Volcano plots of differential expression results for fatal asthma-derived vs. non-asthma-derived ASM A) exposed to control and B) exposed to budesonide. The y-axis corresponds to the negative log (base 10) of Benjamini-Hochberg corrected p-value (q-values) while the x-axis corresponds to the log (base 2) transformed fold change for difference in expression between categories. Differentially expressed transcripts according to a q-value <0.05 are colored in red.

Figure E3. Venn diagram of the significant genes in comparisons of glucocorticoids vs. control in chALL (lime green), ASM (green), and macrophage (orange), LCL (purple), and U2OS (pink). Significant genes in chALL, ASM, and macrophage cells are from integrated differential expression results, while in LCL and U2OS cells are from single study differential expression results. ASM: airway smooth muscle; chALL: childhood acute lymphoblastic leukemia; LEL: lymphoblastoid cell line; U2OS: human bone osteosarcoma epithelial cell line.

Figure E4. *CEBPD* expression changes in response to glucocorticoid exposure across all cell types considered. A) Cell type-specific differential expression results show that *CEBPD* had statistically significant changes in ASM and U2OS cells. B) Sample-specific expression results for all ASM studies show consistent increase of *CEBPD* in response to glucocorticoids, with the smallest fold change observed in fatal asthma donor-derived ASM. ASM: airway smooth muscle; BE: bronchial epithelium; chALL: childhood acute lymphoblastic leukemia; LEL: lymphoblastoid cell line; MCF10A-Myc: immortalized human mammary epithelial cell line MCF10A overexpressing c-Myc; RMA: robust multi-array average; U2OS: human bone osteosarcoma epithelial cell line.

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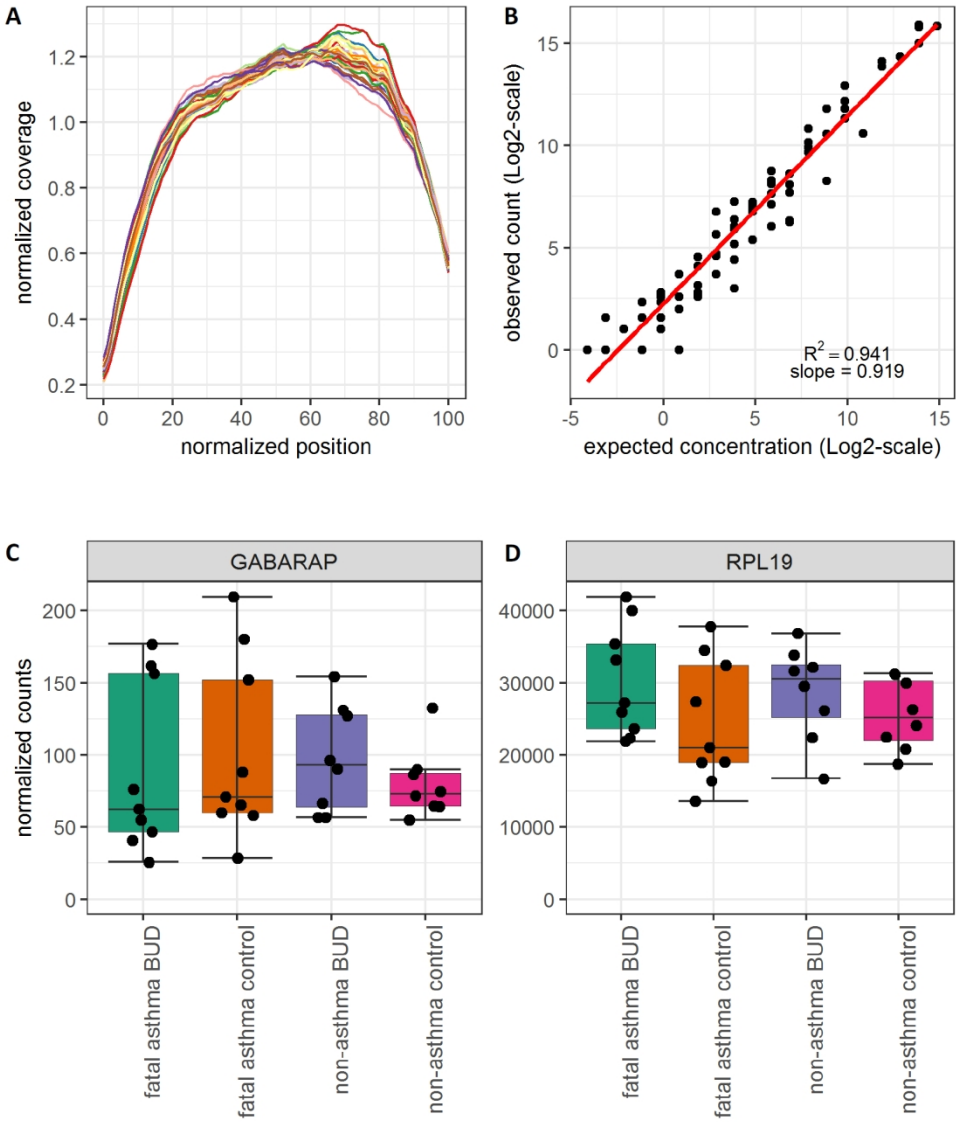


Figure E1

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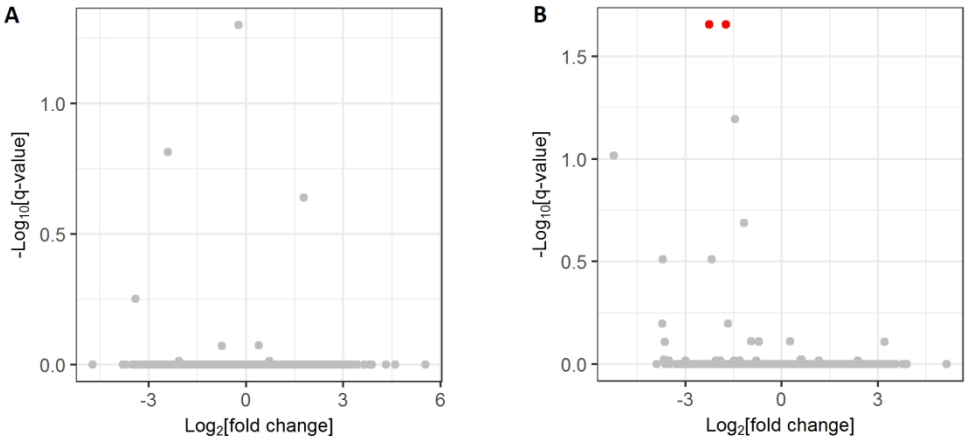


Figure E2

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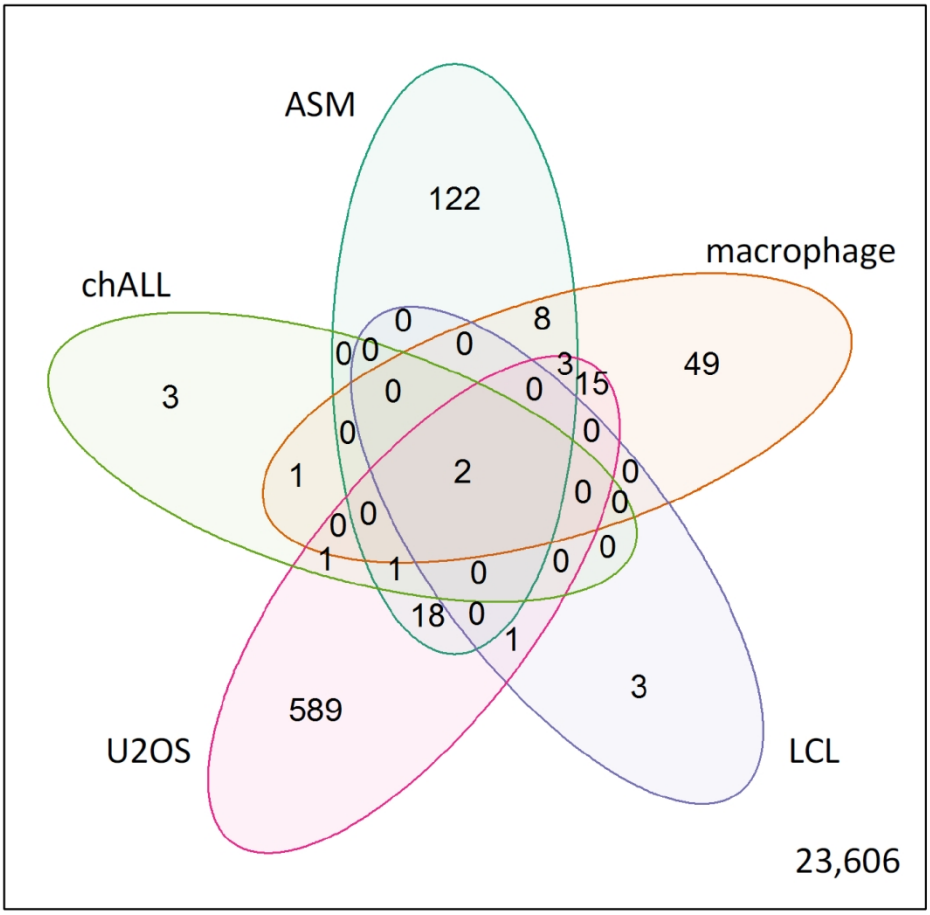


Figure E3

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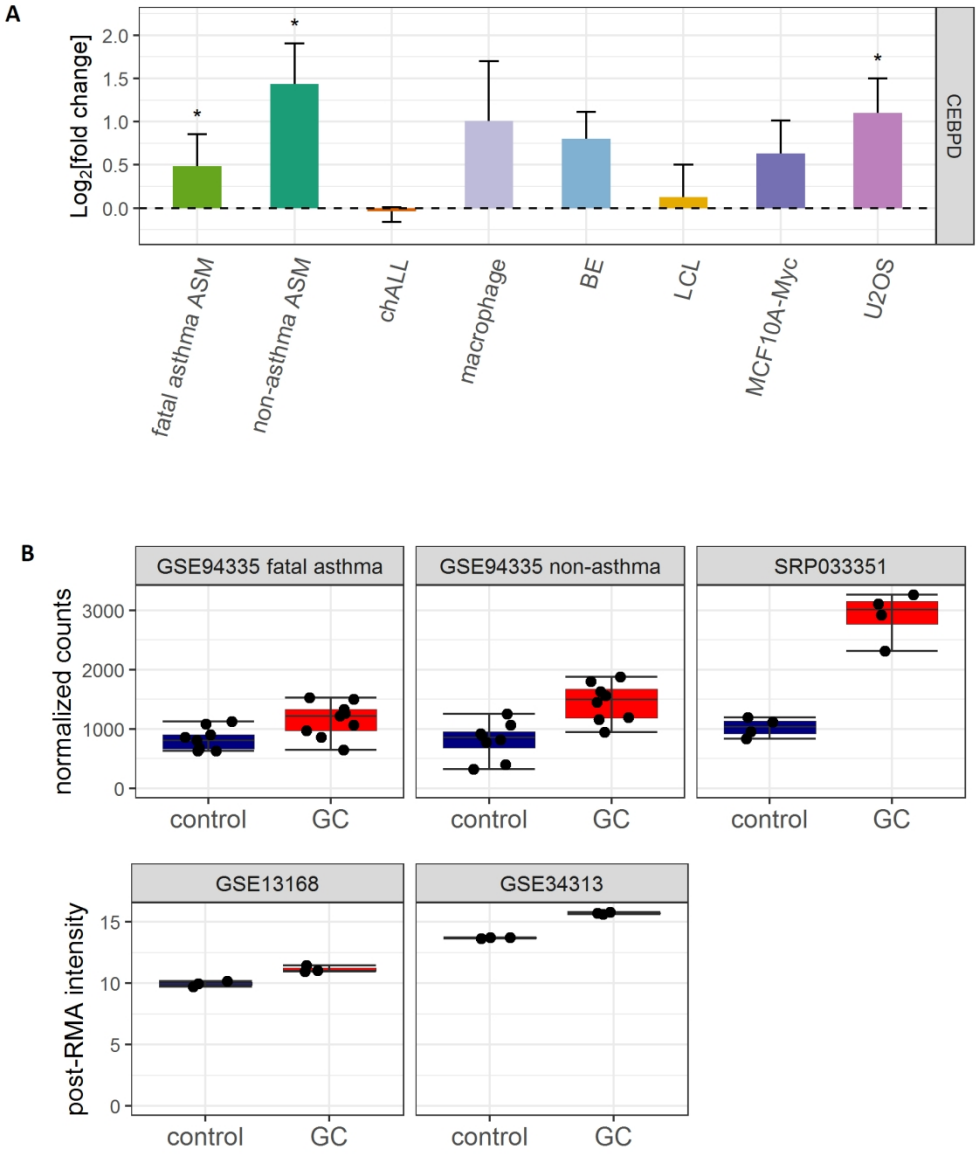


Figure E4

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